

Songtao Liu

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Summary

- Over **20** years of OCP specializing in Oracle databases, and other database and system administration using various administrative tools across major operating systems. Focus on the management of local and cloud-based database systems.
- Proficient in traditional and cloud-based ERP/EMC and SAP system planning, including database creation, configuration, performance tuning, migration, and disaster recovery, with a strong understanding of business processes.
- Highly self-motivated and skilled in teamwork and collaboration, ensuring efficient project management, issue resolution, and timely system transitions.
- Strong leadership and organizational abilities, with experience in ERP system design and project management. A dedicated, team-oriented professional with a proactive approach.

Core Competencies

- Business ERP System Planning and Implementation
- Database Administration
- Oracle Professionals

Professional Experience

Oracle Database Administrator III *Buckeye Telesystem Inc. Toledo, OH* 09/2019 – 12/2024

- Remote management of Oracle/Bentley databases, including Oracle software installation, database creation and configuration, data migration between old and new databases/servers, database backups, performance tuning, DB monitoring and troubleshooting.
- Cloud access (OCI) for Oracle/CSG databases, including the creation and updating of database objects, data retrieval, AI, security, and data testing. Creation and modification of reports for clients and data users.
- Migration of MySQL and PostgreSQL databases, including installation of database software on new servers, database creation and configuration, data migration and database backups, DB monitoring and maintenance.
- Practiced Oracle Data Guard and implemented disaster recovery solutions.
- Completed Oracle/CSG database SSRS report system migration from Vantage to Snowflake.
- SQL Server database and Datawarehouse management, including software installation, database and database object creation and debugging, database object backups and migration, database monitoring, maintenance and troubleshooting.
- Other database-related tasks, including database upgrades, and patching.
- Learned new DBA and IT related technologies and tools such as Oracle Cloud, BigchainDB, HashiCorp/Terraform, AWS DynamoDB/RDS/CloudWatch/CLI, NoSQL (MongoDB, Cassandra, Couchbase, and Redis), Data Lake, TensorFlow, InfluxDB, TimescaleDB, AI Vector Search, and MS Azure, Google Cloud Pub/Sub, Docker, Kubernetes, IaC, Apache Kafka, AWS Kinesis, Prometheus, Grafana, ELK Stack.

Database Administrator *Dynamic Campus Inc. Austin, TX* 10/2018 – 09/2019

- I was employed by the IT management contractor, Dynamic Campus Inc., and continues to fulfill the responsibilities of the Database Administrator position at Lourdes University.

Database Administrator *Lourdes University, Sylvania, OH* 06/2013 – 10/2018

- Physically and logically planned, designed, monitored, and implemented the Banner ERP system, including Oracle databases, MS Sql-Server, data warehouse, Fusion Middleware, WebLogic, and relevant application, VMware data center, and web front-end server environments.
- Oracle and PostgreSQL database installation, configuration, and data migration. Set up database backups on virtual servers and databases for restoring and recovering RDBMS systems.
- Coordinated with other IT management on disaster recovery solutions for all systems and applications.
- Planned and coordinated with internal and external groups of DBAs, Network, Business leaders, functional users, and third-parties to implement projects within multiple systems relationships.
- System upgrade and patch implementation including Oracle databases, clients, security, OS, and applications.
- Migrated databases across platforms from Windows to Linux.
- Troubleshooting, database optimization and performance tuning by using RDBMS performance tuning tools.

- Performed host and database clone, data synchronization, monitoring, load balancing, verifying the integrity and availability of Oracle/Banner ERP system. Ensured the stability and efficient operation of the in-house and hosted information systems that support core organizational functions.

Database Administrator

Albion College, Albion, MI

03/2011 – 06/2013

- Management of Oracle/Banner databases, including Oracle software installation, database creation and configuration, data migration between old and new databases/servers, database backups, performance tuning, DB monitoring and troubleshooting.
- Responsible for the Oracle/Banner database performance and data security of administrative databases housing functions such as student information, course management, financial aid, HR, and advancement.
- Provided quality control and management of internally developed and vendor-provided programs to fulfill end users' data and programming needs.
- Deployed and maintained database security and compliance as required by school policy and standards.
- Introduced and integrated new technologies into existing data center environments.
- Maintained and supported Oracle, PostgreSQL, and SQL server Mirroring, Log shipping, and Real Application Clustering.
- Participated in research projects to evaluate tools, techniques, and upgrades that will further enhance and advance the RDBMS.

System and Database Administrator

Direct Home Décor, Cleveland, OH

02/2004 – 02/2011

- Handled Oracle, SQL Server, and My SQL databases design, installation, configuration, upgrade, optimization, data migration, maintenance, and troubleshooting.
- Installed, configured, and maintained hardware and software, and applied operating system updates, patches, and configuration changes on Windows server environment.
- Analyzed and diagnosed database and system logs and identified potential issues with the database systems and troubleshooting reported problems.
- Responsible for security, license implementation, and documenting the configurations of the database and application systems.
- Interfaced with developers, network, virtualization, storage, and systems operations teams to resolve problems with applications and database systems.
- Coordinated with relevant IT and functional groups to install and configure new hardware, software, and security certificates on the systems of database, application, and front-end web servers.
- Worked with technical team to analyze and diagnose system performance and identify potential issues with management tools and troubleshoot reported problems.
- Periodically reported the project progress and deliverables approaches of the projects to IT, functional groups, and third parties.
- Implemented multiple projects at the same time of business needs to achieve expectations of customers.

Database and Computer Systems Analyst

Shandong Medical University, Jinan, China

07/1993-11/2000

- Designed, installed and implemented Windows servers, and MS Sql Server, Oracle, My Sql, Access, and dBase databases.
- Maintained Windows systems and databases for high availability.
- Performance tuning and SQL optimization on operating systems and databases.
- Performed hardware and software systems support for 180 computers and 110 printers.

Professor and Dean

Shandong Institute of Economics, Jinan, China

07/1984-07/1993

- Directed teaching and management student recruitment, admission, and testing committees.
- Taught Business management strategy, marketing, and small business management.

Certificates and Projects

Certificates

- Oracle Database 19c Administrator Certified Professional
- Oracle Database 11g Administrator Certified Professional
- Oracle Database 10g Administrator Certified Professional
- Oracle Database 9i Administrator Certified Professional
- Oracle Database 9i Administrator Certified Associate
- SunGard Oracle DBA Survival Tips and Tricks training certificate
- SunGard Intro to Banner Administration training certificate
- Microsoft Certified Professional

Projects

- Oracle/CSG database SSRS report system migration from Vantage to Snowflake – 2024

- Tableau server upgrade from 2022.3.1 to 2024.2.3 – 2024
- PostgreSQL database installation, configuration, and data migration – 2016, 2023
- Jira system installation and configuration – 2022
- Oracle/Bentley database creation, configuration, and data migration (19.3) – 2021
- MS Sql Server 2016 implementations – 2019
- Oracle RDBMS and Client upgrades 11.2.0.4 to 12.1.0.2, and 12.2.0.1 – 2018, 2019
- IBM Cognos upgrade project – 2019
- Data Center transformation from Unitrans to Cohesity – 2018 and 2019
- Ellucian Banner/Oracle 9 ERP system implementation – 2018
- Migrated Oracle databases across platforms from Window to Linux – 2018
- MS Sql-Server Recruiter database clone and upgrade project (3.8 to 3.9) – 2016
- Oracle Cognos database components installation on ODS/EDW/SRP/RAP for Cognos reports – 2015
- Higher-Ed report system project – 2015
- Oracle databases creation and configuration on Window Server 2008 – 2015
- Restored/recovered the Oracle Banner production database – 2014
- VMware system upgrade project implementation – 2012

Computer Skills

Database Environments, Tools and Knowledge:

Oracle 8i, 9i, 10g, 11g, 12c, 18c, 19c, 21c, 23ai, Access, MySQL, Oracle Fusion Middleware 11g and 12c, Oracle Application Server, Data Guard, OCI, Oracle EBS, Terraform, Fusion Middleware, WebLogic, Oracle Exadata, Sql Developer 3-22.2, SAC, PerfMon, RMAN, RAC, OMS, GDS, OAM, OEM, GoldenGate, TOAD, ADDM, AWR, ADR, TKProf, SOM, Argos, SSRS, Profiler, DTA, ODBC, JDBC, DBaaS, BigchainDB, Data Lake, Snowflake, TimescaleDB, InfluxDB, AWS DynamoDB/RDS/CloudWatch/CLI, NoSQL(MongoDB, Cassandra, Couchbase, and Redis, TensorFlow, AI Vector Search, MS SQL Server 2005/2008R2/2012/2014/2016/2018/2022, SSMS, SSIS, NetApp

Operating Systems and Knowledge:

Linux (RedHat 6-9, CentOS 7/8/Stream, Fedora 14-35, openSuSe 11-15), Unix (SunOS.Solaris 4-11), Microsoft NT/Windows Servers 2000/2003/2008R2/2012R2/2016R2/2018/2021, XP, Vista, Windows 7, 10, 11

Networking and Tools:

eDirectory, Active Directory, DirXML, MIIS, vCenter, Storage Replication, EMC, HP MSA2000, OLTP, LAN, WAN, ETL (Apache NiFi/Flink, Talend, Informatica, and Kafka Streams), IIS, TDS, TCP/IP, SOAP, FTP, HTTP, Cisco, Gateways, Firewalls, Tableau, Multicloud, AWS, Microsoft Azure, Google Cloud Pub/Sub, Docker, Kubernetes, IaC, Apache Kafka, AWS Kinesis, Prometheus, Grafana, ELK Stack.

Programming Language and Skills:

SQL, PL/SQL, PL/pgSQL, T-SQL, Java, VB, C++, COBOL, PowerShell, Python, .NET, Groove/Gravy, JSON, R, Go.

Education

Master's Degree in Economics

Shandong University

Bachelor's Degree in Information Systems

Shandong University of Finance and Economics

Bachelor's Degree in Material Economy

Shandong University of Finance and Economics